# Install and execute the AFF and FAS firmware Ansible Automation package (Beginners)

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NetApp
March 28, 2024

This PDF was generated from https://docs.netapp.com/us-en/active-iq/task_hosting_firmware_files_using_web_server.html on March 28, 2024. Always check docs.netapp.com for the latest.

# Table of Contents

Install and execute the AFF and FAS firmware Ansible Automation package (Beginners) . . . 1
- Host firmware files using web server . . . 1
- Work with inventory file . . . 1
- Execute Ansible Playbook using NetApp Docker image . . . 3
- Execute Ansible Playbook without NetApp Docker image . . . 4
- Validate firmware installation . . . 5
- Get more information . . . 7

# Install and execute the AFF and FAS firmware Ansible Automation package (Beginners)

## Host firmware files using web server

After you download the automation package, the firmware files should be hosted on a web server.

The web server can be set up in multiple ways. For instructions to set up a simple web server using Python, refer to Webserver using Python.

**Step**

1. Save the base URL of the web server. If the URLs for disk firmware, shelf firmware, and service processor firmware are **http://<web-server>/path/all_shelf_fw.zip**, **http://<web-server>/path/all.zip**, and **http://<web-server>/path/<SP/BMC>_<version_number>_fw.zip**, save **http://<web-server>/path/** as the base URL.

The filename is automatically detected by the Ansible Playbook.

## Work with inventory file

The inventory file consists of the cluster management LIFs of the systems that are eligible for firmware updates. It contains the list of clusters with disk and shelf firmware filename information wherever applicable.

For service processor firmware update, node hostnames and SP/BMC IP is included in the inventory file.

### Inventory file format

The following is a sample inventory file format with both disk and shelf firmware updates:

```
clusters:
  - clustername: <cluster management LIF-1>
    disk_fw_file: all.zip
    shelf_fw_file: all_shelf_fw.zip

  - clustername: <cluster management LIF-2>
    disk_fw_file: all.zip
    sp_nodes:
    - hostname: <node hostname 1>
      sp_fw_file: SP_FW_308-03990_11.5.zip
      sp_fw_type: bmc
      sp_fw_ver: '11.5'
      sp_ip: <BMC IP>
    - hostname: <node hostname 2>
      sp_fw_file: SP_FW_308-03991_5.8.zip
      sp_fw_type: sp
      sp_fw_ver: '5.8'
      sp_ip: <SP IP>
```

In the example, both shelf and disk firmware updates are applicable on cluster-1 and disk and SP/BMC firmware updates are applicable on cluster-2.

## Delete a cluster from the inventory file

In case you do not want to apply firmware updates on a particular cluster, you can remove the cluster from the inventory file.

For example, if you do not want to apply disk firmware updates on cluster-2, you can remove it from the inventory file using the following command:

```
clusters:
  - clustername: <cluster management LIF-1>
    disk_fw_file: all.zip
    shelf_fw_file: all_shelf_fw.zip
```

You can observe that all the data for cluster-2 has been deleted.

If you want to apply only disk firmware updates on cluster-1 and not shelf firmware updates, you can do so using the following command:

```
clusters:
  - clustername: <cluster management LIF-1>
    disk_fw_file: all.zip
```

You can see that the *shelf_fw_file* key and value have been removed from cluster-1.

Manual addition of clusters or controllers is not supported.

# Execute Ansible Playbook using NetApp Docker image

Before you execute the Ansible Playbook, ensure that the **NetApp_Ansible_*.zip** file has been extracted and the web server with disk or shelf firmware files is ready.

## Before you begin

Before executing Ansible Playbook using NetApp docker, you should:

- Download the AFF and FAS firmware Ansible Automation package
- Host the Firmware files using the web server
- Work with the inventory file
- Ensure that NetApp Docker is installed.

### Steps

1. Set up Docker.
2. Pull the NetApp Docker image from DockerHub by executing the following command:

```
$ docker pull schmots1/netapp-ansible

Using default tag: latest
latest: Pulling from schmots1/netapp-ansible
docker.io/schmots1/netapp-ansible:lates
```

   For more information about the docker pull command, refer to the Docker Pull Documentation.

3. Run the Docker image as a container and log in to the container to execute the Ansible Playbook.
4. Copy the path of the folder which contains the extracted Ansible Playbook and inventory files, for example, **downloaded_playbook_path**. The Ansible Playbook and inventory files should be in the same folder for successful execution.
5. Mount the folder as a volume on the Docker container. For example, to mount the folder **container_path**, you should execute the following command:

```
$ docker run -v <downloaded_playbook_path>:/<container_path> -it
schmots1/netapp-ansible:latest /bin/bash
```

   The container starts and the console is now at bash shell of the container. For more information about the Docker Run command, refer to the Docker Run Documentation.

6. Execute the Ansible Playbook inside the container using the **ansible-playbook** command:

```
$ cd <container_path>
$ ansible-playbook na_ontap_pb_upgrade_firmware.yml

Enter your ONTAP admin username: ****
Enter the password for your ONTAP admin user: ****
Enter the base URL to the firmware package (using HTTP is recommended):
http://<web-server>/path/
PLAY [ONTAP Firmware Upgrade]
*****************************************************************
```

> If there are a set of clusters with different login credentials, the Ansible Playbook must be run on each cluster. There are no changes required to the inventory file as the Ansible Playbook skips the clusters for which the login has failed.

For more information about the **ansible-playbook** command, refer to the Ansible Playbook Documentation and to execute the Ansible playbook in check mode (dry run), refer to Ansible: Check mode.

After executing the Ansible Playbook, refer to the Firmware Installation Validations for post-execution instructions.

# Execute Ansible Playbook without NetApp Docker image

**Steps**

1. Install Python and Ansible.
2. Install the required Python packages using **pip**:

```
$ pip install netapp-lib requests paramiko

Installing collected packages: netapp-lib, requests, paramiko
Successfully installed netapp-lib-2020.3.12 requests-2.23.0 paramiko-
2.7.2
```

3. Install NetApp Ansible collection using the **ansible-galaxy** command:

```
To install the collection only for the current user
$ ansible-galaxy collection install netapp.ontap

To do a more universal installation,
$ ansible-galaxy collection install netapp.ontap -p
/usr/share/ansible/collections

$ chmod -R +rw /usr/share/ansible/collections
```

For more information about the ansible-galaxy command, refer to Ansible Galaxy Documentation and for more information about the NetApp Ansible Collection, refer to the NetApp Ansible Collection page.

4. Execute the Ansible Playbook using **ansible-playbook** command:

```
$ cd <downloaded_playbook_path>
$ ansible-playbook na_ontap_pb_upgrade_firmware.yml

Enter your ONTAP admin username: ****
Enter the password for your ONTAP admin user: ****
Enter the base URL to the firmware package (using HTTP is recommended):
http://<web-server>/path/
PLAY [ONTAP Firmware Upgrade]
*****************************************************************
```

> If there are a set of clusters with different login credentials, the Ansible Playbook must be run on each cluster. There are no changes required to the inventory file as the Ansible Playbook skips the clusters for which the login has failed.

For more information about the **ansible-playbook** command, refer to the Ansible Playbook Documentation and to execute the Ansible Playbook in check mode (dry run), refer to Ansible: Check mode.

After executing the playbook, refer to the Firmware Installation Validations for post-execution instructions.

# Validate firmware installation

After the execution of the playbook, log in to the cluster as the cluster administrator.

## Validate disk firmware installation

**Steps**

1. Verify that the drive firmware is installed:

```
::*> storage disk show -fields firmware-revision,model
disk     firmware-revision model
--------------- ----------------- ----------------
1.11.0   NA01              X423_HCOBE900A10
1.11.1   NA01              X423_HCOBE900A10
1.11.2   NA01              X423_HCOBE900A10
1.11.3   NA01              X423_HCOBE900A10
1.11.4   NA01              X423_HCOBE900A10
```

For more information about the command, refer to storage disk show.

2. Verify that the new NVMe Flash Cache firmware is installed:

```
::*> system controller flash-cache show
```

For more information about the command, refer to system controller flash-cache show.

## Validate shelf firmware installation

**Steps**

1. Verify that the new shelf firmware is updated:

   ```
   ::*> system node run -node * -command sysconfig -v
   ```

   In the output, verify that each shelf's firmware is updated to the desired level. For example:

   ```
   Shelf 1: IOM6 Firmware rev. IOM6 A: 0191 IOM3 B: 0191
   ```

   For more information about the command, refer to system node run.

2. Verify that the new ACP firmware is updated:

   ```
   ::*> storage shelf acp module show -instance
   ```

   For more information about the command, refer to storage shelf acp module show.

3. Verify that the desired ACP mode is configured:

   ```
   ::*> storage shelf acp show
   ```

   For more information about the command, refer to storage shelf acp show.

4. Change the ACP mode (channel):

   ```
   ::*> storage shelf acp configure -channel [in-band | out-of-band]
   ```

   For more information about the command, refer to storage shelf acp configure.

## Validating SP/BMC Firmware installation

The Ansible Playbook for Service Processor/BMC firmware updates is enabled with an option to verify the installation of latest SP/BMC firmware on the controller. After the verification is complete (the updates could take a maximum time of two hours), the Ansible Playbook applies internal switch firmware updates by connecting to the SP/BMC console.

The failure and success information for SP/BMC firmware and the internal switch firmware installations will be notified at the end of Ansible Playbook execution. Follow the steps mentioned in the Ansible Playbook in case the SP/BMC firmware/internal switch firmware installation fails.

# Get more information

You can get help and find more information through various resources.

- Troubleshooting information
- Slack workspace
- Email
- Support button in Digital Advisor for support and feedback.

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